Data collection campaign: building a smell cartography

Cartography of Grenoble, Saint Martin d'Hères and Echirolles

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Introduction:

This document contains the answer of the Exercise 2 “Data collection campaign: building a smell cartography”

Task:

Build a table (four columns) specifying the type of data to collect (1) with their associated V’s (2) and discuss how they can contributed to build the smell identity of a street, an area, a quarter, and the whole city (3). Finally, in column 4, specify one or several strategies to collect data. You can also search on internet whether there are data collections available that can contribute to provide insight to address the research question.

Table:

Type of data	**Relevant V’s of Big Data**	**Contribution to smell identity (street / area / quarter / city)**	**Data collection strategy**
Citizen-reported smell observations	Variety, Volume, Veracity, Value	Captures how people *actually* experience the city’s smells, from good to bad.	Create a small app or form where citizens can rate and describe smells with their location and time of the day.
Social media posts mentioning odors	Volume, Velocity, Variety	Real-time feedback, sometimes emotional, about what smells people notice.	Use APIs to collect public posts (Twitter, Instagram) with smell-related words and geolocation.
Air quality sensor data (NO_2, SO_2, PM2.5, VOCs)	Volume, Velocity, Veracity, Value	Measures pollutants that are usually related to industrial or car smells.	Use data from Atmo Auvergne-Rhône-Alpes and install extra low-cost sensors if needed.
Traffic flow and vehicle type	Volume, Velocity, Value	Indicates car-related smells (fuel, exhaust).	Use smart traffic data and GPS mobility sources.
Street cleaning frequency	Veracity, Value	Clean streets smell better; dirtier ones accumulate waste odors.	Get cleaning schedules from local services.
Trash collection routes and frequency	Volume, Veracity, Value	Affects how long trash stays in the street before pickup, which increases the presence of bad odors there.	Obtain municipal waste collection data and routes.
Distance to trash depots / waste treatment centers	Variety, Value	Odors may spread from waste depots to nearby areas depending on wind and distance.	Use GIS maps to calculate proximity.

Type of data	Relevant V's of Big Data	Contribution to smell identity (street / area / quarter / city)	Data collection strategy
Distance from factories / industrial zones	Variety, Value	Industrial smells travel depending on proximity and wind direction.	Use Google maps.
Cafeterias, factories, restaurants (and their waste practices)	Variety, Veracity, Value	Identifies where food or industrial smells come from.	Get data from city business registries or public inspections.
Number of pets / stray animals	Variety, Veracity, Value	Animals also contribute to organic smells in streets or parks through their fur/skin/etc and waste	Use municipal animal control data and citizen reports.
Tree and vegetation density	Variety, Value	Natural odors and air freshness increase with greenery.	Use satellite imagery or city green-space data.
Rain frequency and intensity	Velocity, Value	Rain "cleans" the air but also changes how long odors stay around.	Combine historical and real-time weather data.
Weather data (wind, humidity, temperature)	Velocity, Variety, Value	Weather changes how smells move and how strong they feel.	Use Météo France API or open-source data.

Discussion:

To build a smell cartography for Grenoble, Saint-Martin-d'Hères and Échirolles, it's important to mix different kinds of data. Citizen participation is vital to achieve this goal, as it allows the people to report what they smell, making their presence valued in this exercise. However, that is not enough to cover the smell cartography, as multiple other sources are needed to finish drawing the map. After all, in order to have a complete idea about the smell, finding its source and how it spreads can complete the missing clues about its true identity. Things such as Weather data, Rain frequency and intensity, how strict and often the cleaning is, air quality sensors, types of building in the city (cafeterias/factories/...) , density of trees and green spaces, and number of pets and stray animal can provide a better "image" of the smell, an "image" that can not be felt through the human nose. This finding can show the reasons behind the good smells and the bad ones along with detecting the problem at its core.

- **Weather and rain** play a very important role in how odors travel and change. Wind direction can direct smells from one place to another, humidity can strengthen them, and temperature affects their duration. Rain can "wash" the city temporarily, washing away particles and dust that hold smells, but sometimes after a rainy period, organic odors like soil, mold, or waste become more noticeable, due to being mixed with water. This shows how the smell identity of an area can vary from time to time and place to place
- **Green spaces, trees, and animals** add another natural dimension to the smell map. Trees and parks give off floral, earthy, or wood-like scents that usually make areas smell fresher. At the same time, they filter pollution and balance unpleasant odors from traffic or waste. However, areas with many stray animals or poorly maintained green zones can also produce organic smells that affect the local experience. Including data about vegetation and animal presence helps understand both the positive and negative contributions of nature to the city's smellscape. However, even green spaces can give bad smells when uncared for, as such moderate care is needed to prevent that
- **Factories, cafés, and traffic flow** represent the human and industrial side of the equation. Factories or industrial zones can emit strong chemical odors, this is especially noticed when close to the industrial zone. Cafés and restaurants add pleasant smells of food and coffee, giving the street its cozy smell. Traffic, on the other hand, brings exhaust fumes and fuel odors that dominate main roads during rush hours, this is especially noticed when cars are not maintained or

malfunctionning. Analyzing these activities together shows how urban life creates constant changes in the smell environment depending on the time of day and the type of area. This is why many governments encourage the use of public transport and especially bikes as their use will slowly but surely decrease CO_2 emission along with other gases. This is proven during the corona outbreak

- **Waste management** is another major factor that shapes how a city smells. The frequency of trash collection, the condition of bins, and the distance to waste depots or treatment centers all affect odor intensity. If bins are not emptied often enough or cleaning is irregular, the smell of waste becomes persistent. On the other hand, organized waste routes and strict cleaning schedules reduce these effects. Understanding these patterns helps the city locate where odors come from and act before they spread further. Active participation from the citizens can optimize the frequency of cleaning and places of the trash bin, and this is through the use of online surveys to get their opinion on how good the waste management is.

When combining all this, the five V's of Big Data become visible:

- Volume: big amounts of citizen input, sensor data, and weather records;
- Velocity: live data streams from sensors and social media;
- Variety: very different data formats (text, GPS, weather…);
- Veracity: cross-checking citizen perception with scientific data;
- Value: using the map for public health, urban planning, and quality-of-life improvements.

The approach stays neutral and inclusive, avoiding stereotypes or cultural bias about where "good" or "bad" smells come from. Instead, every zone is analysed objectively. The goal isn't to blame or glorify any group, but to understand how infrastructure, environment, and people together shape the smell identity of the city.